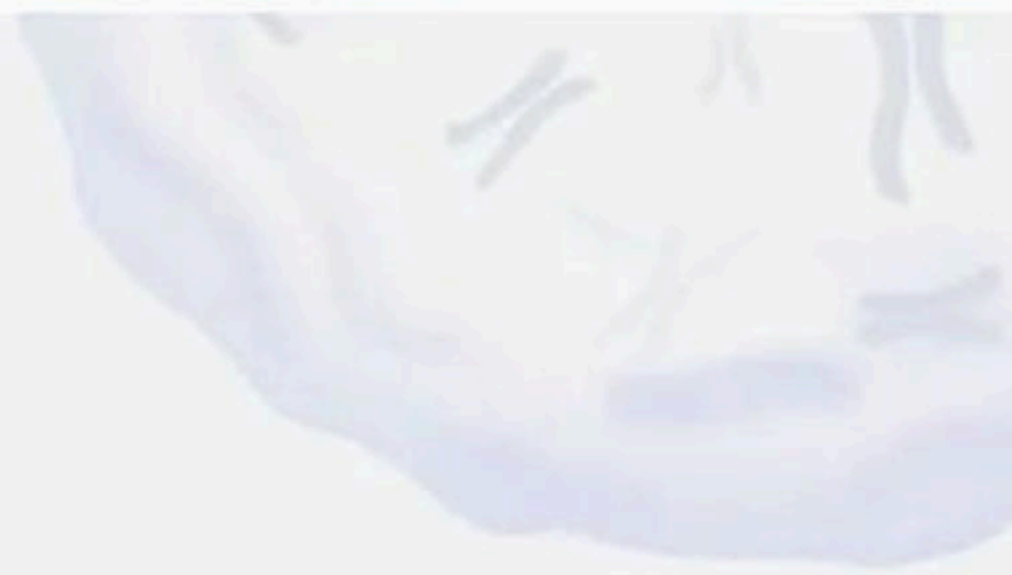
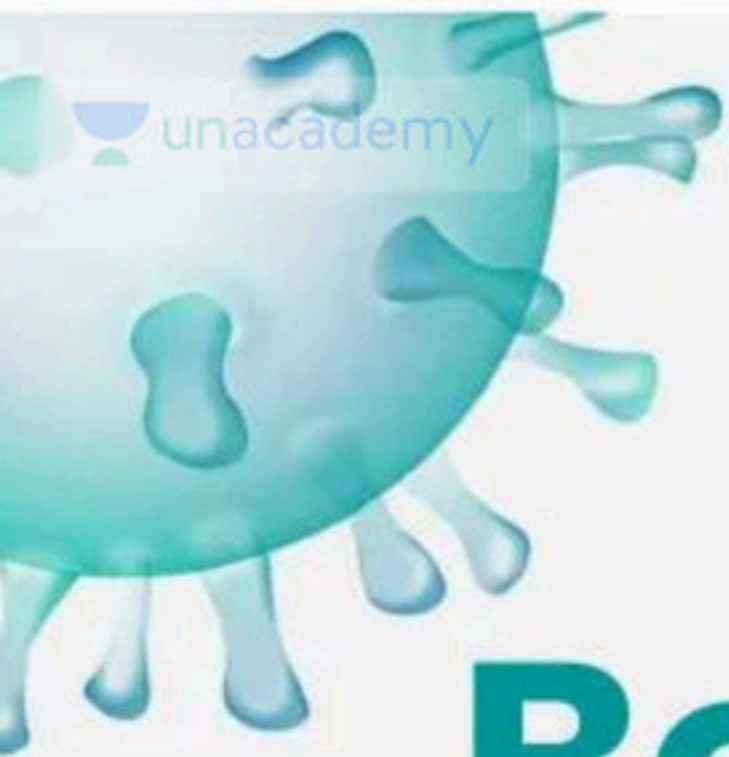
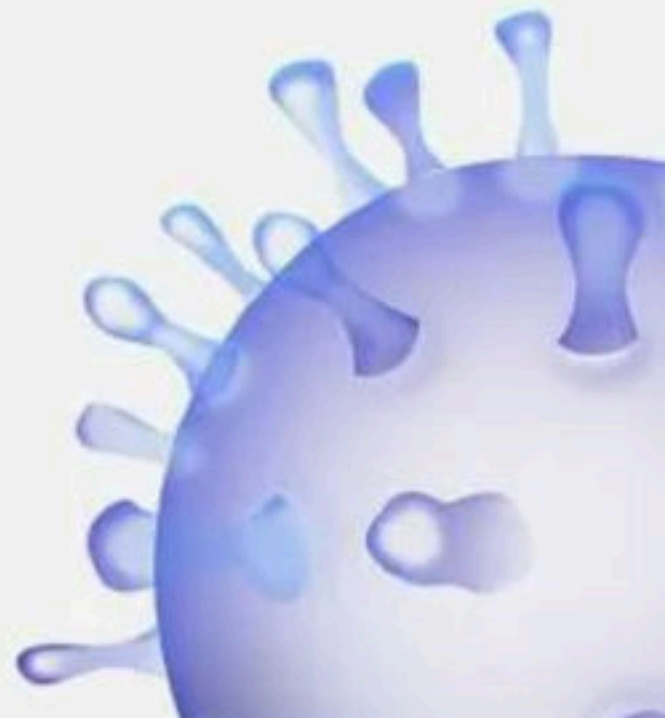


Body Fluids and Circulation - 100 MCQs

NCERT Nichod : Biology Practice Series NEET 2025



Body Fluids and Circulation



BIOLOGYLIVE

1. SA node is called the pace maker of the heart. Why?

- (1) It initiates and maintains the rhythmic contractile activity of heart.
- (2) It delays the transmission of impulse between the atria and ventricles.
- (3) It gets stimulated when it receives neural signal.
- (4) It maintains the pace of the heart beat



2. Select the correct Rh-blood groups of the parents, whose child is affected with erythroblastosis fetalis.

- (1) Both father and mother are Rh + ve
- (2) Mother is Rh + ve and father is Rh-ve
- (3) Both father and mother are Rh-ve
- (4) Father is Rh + ve and mother is Rh – ve



3. Match the columns I and II and choose the correct option.

Column I

- A. Coronary artery disease
- B. Angina pectoris
- C. Symptoms of high bloodpressure
- D. Congestive heart failure

Column II

- P. Affects brain and kidney
- Q. Caused by deposits of calcium, fat, cholesterol which makes the lumen of arteries narrower
- R. Lungs mainly affected in this disease
- S. Acute chest pain appears when no enough oxygen is reaching the heart muscle

(1) A-Q, B-S, C-P, D-R

(2) A-Q, B-S, C-R, D-P

(3) A-S, B-Q, C-P, D-R

(4) A-P, B-R, C-S, D-Q



4. Match the columns I and II and choose the correct option.

(1) A-Q, B-R, C-P

(2) A-Q, B-P, C-R

(3) A-R, B-Q, C-P

(4) A-P, B-R, C-Q

Column I		Column II	
A	Heart failure	P	When heart stops beating
B	Cardiac arrest	Q	When heart is not pumping blood effectively enough to meet the needs of the body
C	Heart attack	R	When the heart muscle is suddenly damaged by an inadequate blood supply



5. Match the columns I and II and choose the correct option.

Column I

A. Neutrophils

B. Basophils

C. Monocytes

D. Eosinophils

Column II

P. 0.5-1%

Q. 60-65%

R. 2-3%

S. 6-8%

(1) A-P, B-Q, C-S, D-R

(2) A-Q, B-P, C-S, D-R

(3) A-S, B-R, C-Q, D-P

(4) A-S, B-R, C-P, D-Q



6. Four chambered heart is found in

- (1) Turtle
- (2) Tortoise
- (3) 1 and 2
- (4) Crocodile



7. The QRS complex in a standard ECG represents:

- (1) Depolarisation of auricles
- (2) Depolarisation of ventricles
- (3) Repolarisation of ventricles
- (4) Repolarisation of auricles



8. Which one of the following is absent in the human beings?

- (1) Nucleated RBC
- (2) Conus arteriosus
- (3) Renal portal vein
- (4) All of them



9. Cardiac output is

- (1) Stroke volume $\times$ Heart rate = 5 L/m
- (2) Stroke volume $\times$ Heart rate = 500ml
- (3) Stroke volume $\times$ Heart rate = 72ml/m
- (4) Stroke volume $\times$ Heart rate = 50 L/m



10. Neural signal through the sympathetic nervous (part of ANS) increases the cardiac output because of

- (1) Increasing the rate of heartbeat
- (2) Increasing the strength of atrial contraction
- (3) 1 and 2
- (4) Decreasing the speed of conduction



11. The first heart sound is caused by the closure of the valves.

- (1) Tricuspid and bicuspid- semilunar valve
- (2) Pulmonary semilunar
- (3) Mitral
- (4) Tricuspid and bicuspid-AV valve



12. The typical 'lub-dub' sounds heard during heartbeat are produced due to

- (1) Closure of semilunar valves
- (2) Closure of bicuspid and tricuspid valves
- (3) Closure of bicuspid and tricuspid valves followed by semilunar valves
- (4) Opening of bicuspid and tricuspid valves followed by semilunar valves



13. The formed element constitutes how much percent of blood?

- (1) 55
- (2) 45
- (3) 6-8
- (4) 90-92



14. The right atrium and the right ventricle are separated by a ____ tissue called the ____ septum

- (1) Thick fibrous, ventricular
- (2) Thick fibrous, atrio-ventricular
- (3) Thin fibrous, atrio-ventricular
- (4) Thick elastic, atrio-ventricular



15. Parasympathetic neural signal decreases the cardiac output by

- (1) Decreasing the rate of heartbeat
- (2) Decreasing the speed conduction of action potential
- (3) 1 and 2
- (4) Increasing heart rate



16. Electrocardiogram is a

- (1) Artificial heart
- (2) Process
- (3) Machine
- (4) Graph



17. Which of the following is correct?

- (1) Plasma - Blood
- (2) Atrioventricular node - Pacemaker
- (3) Leucocytes - Lymphocytes
- (4) Mitral valve - Bicuspid valve



18. A vein that collects blood from all the parts of the body _____

- (1) Pulmonary vein
- (2) Coronary vein
- (3) Carotid vein
- (4) Vena cava



19. A special neural centre in the _____ the cardiac function through _____ can moderate

- (1) Medulla oblongata, autonomic nervous system (ANS)
- (2) Pons, autonomic nervous system (ANS)
- (3) Medulla oblongata, sympathetic nervous system only
- (4) Medulla oblongata, parasympathetic nervous system only



20. Statement A : Group O blood can be donated to persons with any other blood group and hence O group individuals are called universal recipient.

Statement B : Person with AB group can accept blood from persons with AB as well as other groups of blood. Such persons are called universal donor.

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



21. Identify the correct statement regarding cardiac activity:

- I. Normal activities of the human heart is regulated intrinsically, hence it is neurogenic.**
- II. A special neural centre in the medulla oblongata can moderate the cardiac function through the CNS**
- III. Parasympathetic neural signals increase the rate of heart beat**
- IV. Adrenal medullary hormones can increase cardiac output.**

(1) IV only

(2) IV and I

(3) II only

(4) II and III



22. Identify the correct statement

I. The oxygenated blood is emptied into the right atrium.

II. The hepatic portal vein carries blood from intestine to the liver before it is delivered to the systemic circulation.

(1) I is correct

(2) II is correct

(3) Both are correct

(4) Both are incorrect



23. Statement A : Fibrinogens primarily are involved in defense mechanisms of the body

Statement B : Clotting factors are present in the plasma in an inactive form.

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



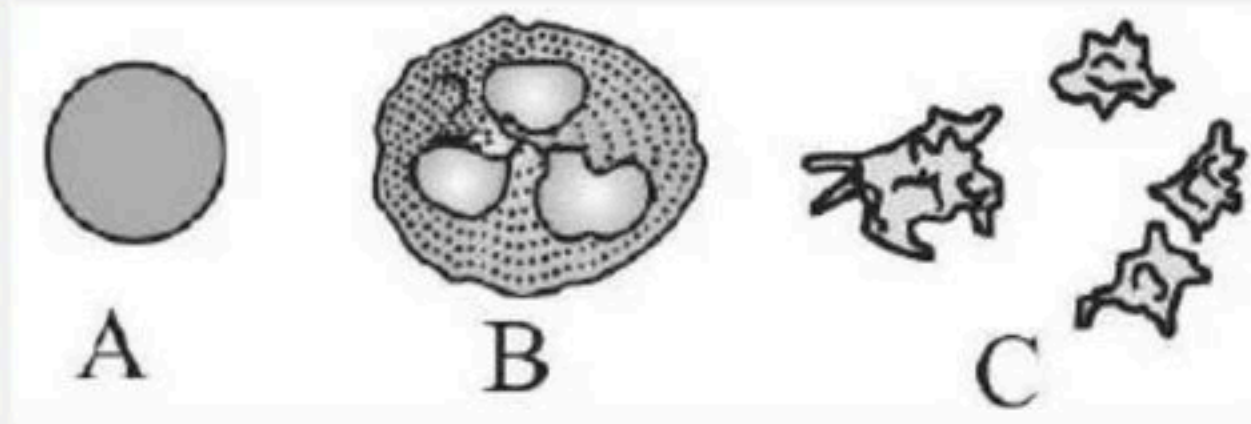
24. Statement A : The opening between left atrium and left ventricle is guarded by mitral valve.

Statement B : The opening of the right and left ventricles into the pulmonary artery and the aorta respectively are provided with the semi-lunar valves.

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



25. The following diagram shows different types of organisms (A-C). Identify the correct option w.r.t their number?



(1) A-6000-8000 mm³, B – 1, 5000, 00 – 3, 500, 00 mm³, C-5 millions - 5.5 million mm³

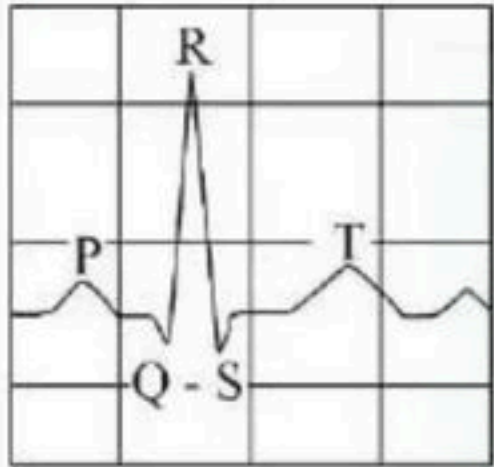
(2) A-1,5000,00-3,500,00 mm³, B-5 millions to 5.5 million mm³, C- 6000 – 8000 mm³

(3) A-5 millions to 5.5 million mm³, B-6000-8000 mm³, C- 1,50,000-3,00,000 mm³

(4) A-5 millions to 5.5 million mm³, B- 6000 – 8000 mm³, C- 1,5000,003, 500, 00 mm³



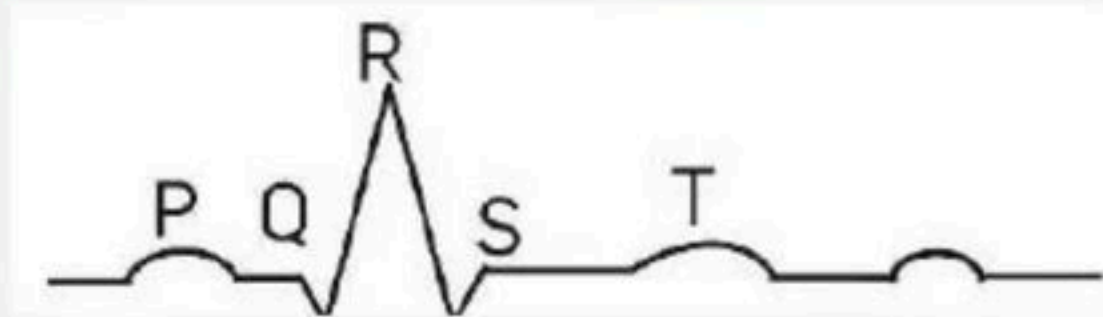
26. The diagram given here is the standard ECG of a normal person. The P-wave represents the



- (1) Beginning of the systole
- (2) End of systole
- (3) Contraction of both the atria
- (4) Initiation of the ventricular contraction



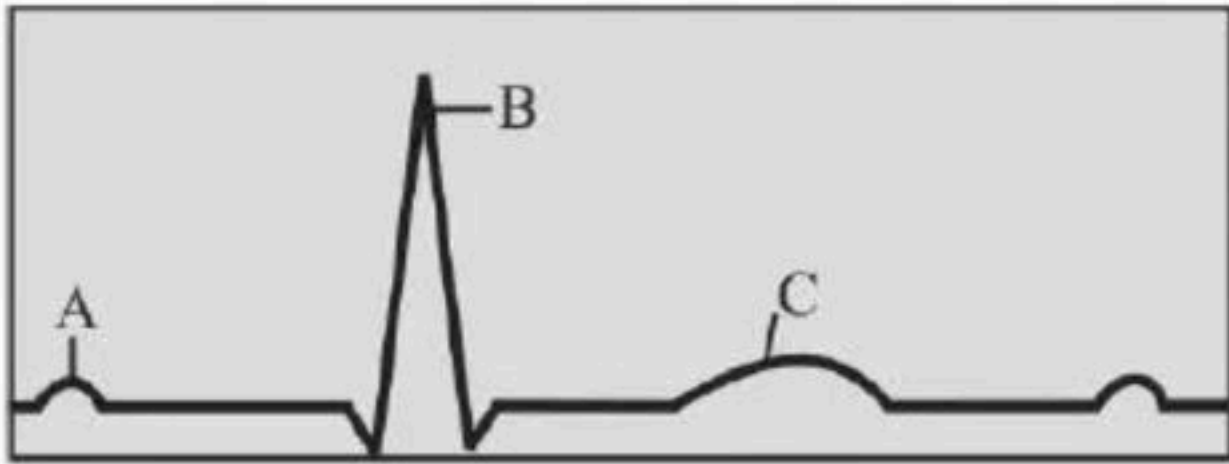
27. The given figure represents diagrammatic presentation of an ECG. Each peak in the ECG is identified with a letter from P to T that corresponds to a specific electrical activity of the heart.



- (1) QRS - Depolarization of the ventricles
- (2) R wave - Marks the end of the systole.
- (3) P-Electrical excitation (or depolarization) of the atria
- (4) T wave - Return of the ventricles from excited to normal state (repolarization).



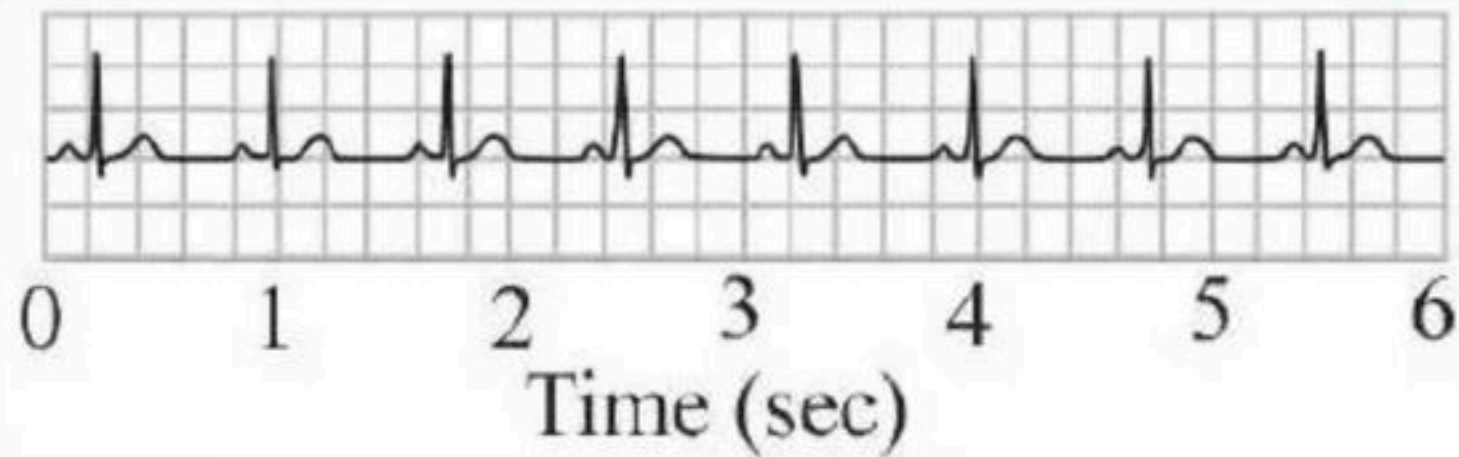
28. The following figure shows graphical representation of ECG. Identify the waves of ECG (A-C)-



- (1) A- QRS complex, B- T-wave, C- P-wave
- (2) A- QRS complex, B- P-wave, C- T-wave
- (3) A- P-wave, B- QRS complex, C- T-wave
- (4) A- T-wave, B- P-wave, C- QRS complex



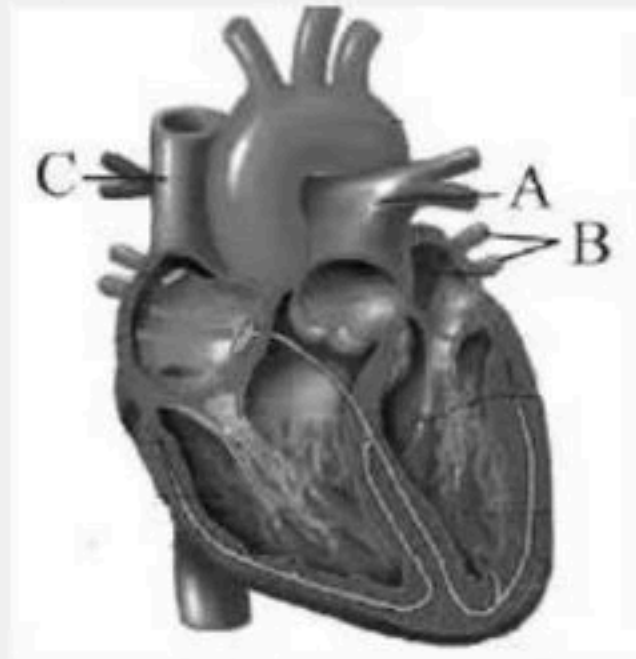
29. What is the heart rate in the following ECG?



- (1) 80
- (2) 75
- (3) 85
- (4) 70



30. The deoxygenated blood from heart carried by _____ to lungs and oxygenated blood from lungs pumped into heart by _____ respectively?



(1) B, A

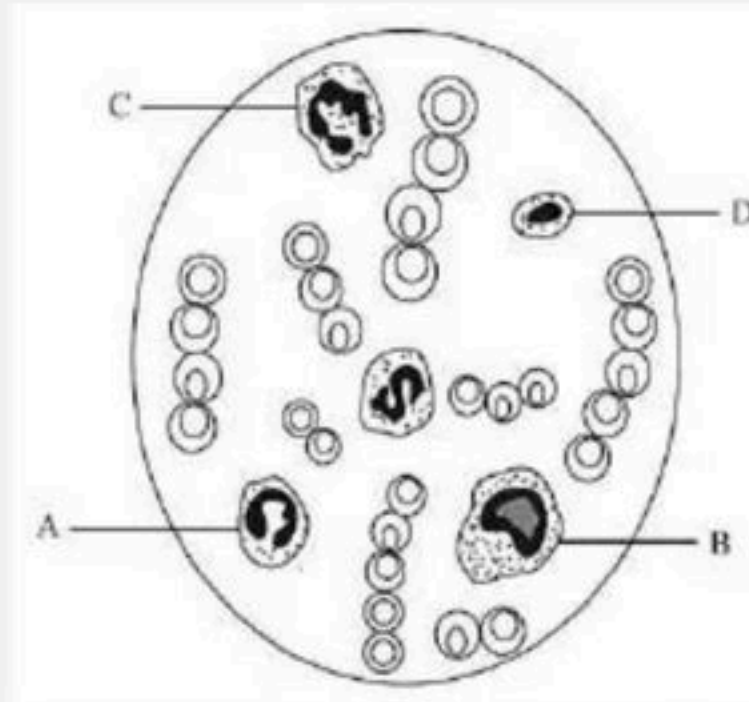
(2) A, B

(3) C, A

(4) A, B



31. Study the diagram given below and identify the cells labelled as A, B, C and D, and choose the correct option.



- (1) A - Eosinophil, B - Erythrocyte, C - Neutrophil and D - Basophil
- (2) A-Eosinophil, B- Lymphocyte, C-Neutrophil and D - Monocyte.
- (3) A - Erythrocyte, B - Basophil, C - Neutrophil and D - Lymphocyte
- (4) A - Eosinophil, B - Monocyte, C - Neutrophil and D - Lymphocyte



32. Assertion : Left ventricle of heart has thickest wall than that of right ventricle.

Reason : Right ventricle need to pump blood to nearby lungs only

- (1) Both Assertion and Reason are correct and Reason is the correct explanation of the Assertion.
- (2) Both Assertion and Reason are correct but Reason is not the correct explanation of the Assertion.
- (3) Assertion is correct but Reason is incorrect.
- (4) Both Assertion and Reason are incorrect



33. Assertion : Open circulatory system is found in most arthropods.

Reason : Arthropods contain haemolymph which directly bathes internal tissues and organs.

- (1) Both Assertion and Reason are correct and Reason is the correct explanation of the Assertion.
- (2) Both Assertion and Reason are correct but Reason is not the correct explanation of the Assertion.
- (3) Assertion is correct but Reason is incorrect.
- (4) Both Assertion and Reason are incorrect



34. Assertion : Atherosclerosis is a disease characterized by the thickening of arterial walls.

Reason : Deposition of cholesterol and triglycerides in the arterial walls cause atherosclerosis.

- (1) Both Assertion and Reason are correct and Reason is the correct explanation of the Assertion.
- (2) Both Assertion and Reason are correct but Reason is not the correct explanation of the Assertion.
- (3) Assertion is correct but Reason is incorrect.
- (4) Both Assertion and Reason are incorrect



35. Assertion : Human heart is myogenic, as it generates its own electrochemical impulse.

Reason : It is not under any neural or endocrine control.

- (1) Both Assertion and Reason are correct and Reason is the correct explanation of the Assertion.
- (2) Both Assertion and Reason are correct but Reason is not the correct explanation of the Assertion.
- (3) Assertion is correct but Reason is incorrect.
- (4) Both Assertion and Reason are incorrect



36. Assertion : Arteriosclerosis is known as hardening of arteries.

Reason : The arterial walls become thick and inelastic due to deposition of cholesterol and calcium salts.

- (1) Both Assertion and Reason are correct and Reason is the correct explanation of the Assertion.
- (2) Both Assertion and Reason are correct but Reason is not the correct explanation of the Assertion.
- (3) Assertion is correct but Reason is incorrect.
- (4) Both Assertion and Reason are incorrect



37. Assertion : Blood platelets play a very important role in coagulation of blood.

Reason : When blood is shed, the platelets disintegrate and liberate thromboplastin which activates prothrombin into thrombin.

- (1) Both Assertion and Reason are correct and Reason is the correct explanation of the Assertion.
- (2) Both Assertion and Reason are correct but Reason is not the correct explanation of the Assertion.
- (3) Assertion is correct but Reason is incorrect.
- (4) Both Assertion and Reason are incorrect



38. Match the following columns and select the Correct option.

Column I

A. Eosinophils

B. Basophils

C. Neutrophils

D. Lymphocytes

Column II

P. Immune response

Q. Phagocytosis

R. Release histaminase, destructive enzymes

S. Release granules containing histamine

(1) A - S, B - P, C - Q, D - R

(2) A - P, B - Q, C - S, D - R

(3) A - Q, B - P, C - R, D - S

(4) A - R, B - S, C - Q, D - P



39. Match the Column-I with Column-II Select the correct option.

Column I

A. P-wave

B. QRS complex

C. T-wave

D. Reduction in the size of T-wave

Column II

P. Depolarisation of ventricles

Q. Repolarisation of ventricles

R. Coronary ischemia

S. Depolarisation of atria

T. Repolarisation of atria

(1) A - S, B - P, C - Q, D -R

(3) A - Q, B - P, C - T, D -R

(2) A - S, B - P, C - Q, D -T

(4) A - Q, B - R, C - T, D -S



40. Match the columns I and II and choose the correct option.

Column I

A. Systemic circulation

B. Hepatic portal system

C. Hepatic portal vein

D. Coronary system

(1) A-Q, B-S, C-P, D-R

(3) A-S, B-Q, C-P, D-R

Column II

P. Circulation of blood to and from the cardiac musculature

Q. Provides nutrients, O_2 to the tissues

R. Carries blood from intestine to the liver

S. Vascular connection exists between digestive tract and liver

(2) A-Q, B-S, C-R, D-P

(4) A-P, B-R, C-S, D-Q



41. Match the columns I and II and choose the correct option.

Column I

- A. Thrombin
- B. Thrombokinase
- C. Calcium
- D. Brown scum

- (1) A-Q, B-R, C-S, D-P
- (2) A-Q, B-R, C-P, D-S
- (3) A-R, B-Q, C-S, D-P
- (4) A-R, B-P, C-Q, D-S

Column II

- P. Conversion of prothrombin to thrombin
- Q. Important role in blood clotting
- R. Conversion of fibrinogen to fibrin
- S. Clot formed mainly of a network of threads called fibrins



42. Match the columns I and II and choose the correct option.

Column I

- A. Blood group- A
- B. Blood group-B
- C. Blood group- AB
- D. Blood group- O

Column II

- P. Universal donor
- Q. Anti-A antibodies in plasma
- R. Universal recipient
- S. Anti-B antibodies in plasma

- (1) A-S, B-Q, C-R, D-P
- (2) A-R, B-P, C-Q, D-S
- (3) A-R, B-Q, C-P, D-S
- (4) A-P, B-Q, C-R, D-S



43. Match the columns I and II and choose the correct option.

Column I		Column II	
A	SA node	P	Minute fibres present throughout the ventricular muscles
B	AV node	Q	Lower left corner of the right atrium
C	Bundle of His	R	Right upper corner of right atrium
D	Purkinje fibres	S	Fibres alongwith right and left bundle

- (1) A-R, B-S, C-Q, D-P
- (2) A-R, B-Q, C-S, D-P
- (3) A-Q, B-R, C-S, D-P
- (4) A-Q, B-P, C-S, D-R



44. Match the columns I and II and choose the correct option.

Column I

A. Globulins

B. Fibrinogen

C. Albumin

(1) A-P, B-R, C-Q

(2) A-Q, B-R, C-P

(3) A-P, B-Q, C-R

(4) A-R, B-Q, C-P

Column II

P. Osmotic balance

Q. Involved in defense mechanisms of the body

R. Blood coagulation



45. Match the columns I and II and choose the correct option.

Column I

- A. Blood
- B. Lymph
- C. Plasma
- D. Serum

Column II

- P. Tissue fluid
- Q. Fluid connective tissue
- R. Plasma without the clotting factors
- S. Straw coloured, viscous fluid

- (1) A-P, B-R, C-Q, D-S
- (2) A-Q, B-P, C-S, D-R
- (3) A-Q, B-S, C-R, D-P
- (4) A-P, B-Q, C-R, D-S



46. Match the columns I and II and choose the correct option.

Column I

- A. Open circulatory system
- B. Closed circulatory system
- C. 2-chambered heart
- D. 3-chambered heart

- (1) A-R, B-P, C-S, D-Q
- (2) A-Q, B-P, C-R, D-S
- (3) A-Q, B-P, C-S, D-R
- (4) A-R, B-P, C-Q, D-S

Column II

- P. Annelids
- Q. Arthropods
- R. Pisces
- S. Reptilia



47. Match the columns I and II and choose the correct option.

Column I

A. Red blood cells

B. Haemoglobin

C. Spleen

D. White blood cells

(1) A-Q, B-P, C-Q, D-R

(2) A-Q, B-P, C-R, D-P

(3) A-P, B-R, C-Q, D-R

(4) A-R, B-P, C-S, D-Q

Column II

P. Transport of respiratory gases

Q. Generally short lived

R. Biconcave in shape and average life span 120 days

S. Graveyard of RBCs



48. Match the columns I and II and choose the correct option.

Column I		Column II	
A	Heart beat	P	5 L
B	Cardiac cycle	Q	70 ml
C	Stroke volume	R	0.8 sec
D	Cardiac output	S	72 per minute

- (1) A-P, B-R, C-Q, D-S
- (2) A-Q, B-P, C-R, D-S
- (3) A-Q, B-R, C-S, D-P
- (4) A-S, B-R, C-Q, D-P



49. Match the columns I and II and choose the correct option.

Column I

A. Heart

B. Pericardium

C. Atria

D. Ventricles

(1) A-S, B-P, C-Q, D-R

(2) A-P, B-S, C-Q, D-R

(3) A-Q, B-S, C-P, D-R

(4) A-S, B-Q, C-P, D-R

Column II

P. Double walled membranous bag protecting heart

Q. Small upper chambers of heart

R. Larger lower chambers of heart

S. Mesoderm in origin



50. Match the columns I and II and choose the correct option.

Column I

- A. Heart beat**
- B. Cardiac output**
- C. Cardiac cycle**

- (1) A-P, B-R, C-Q
- (2) A-Q, B-R, C-P
- (3) A-R, B-P, C-Q
- (4) A-Q, B-P, C-R

Column II

- P. Consists of systole and diastole of both atria and ventricles**
- Q. Many cardiac cycles performed per minute**
- R. Stroke volume multiplied by heart rate**



51. Match the columns I and II and choose the correct option.

Column I

A. ECG

B. P-wave

C. QRS complex

D. T- wave

Column II

P. Depolarisation of the ventricles

Q. Representation of electrical activity of heart during cardiac cycle

R. Repolarisation of the ventricles

S. Depolarisation of the atria

(1) A-Q, B-S, C-P, D-R

(2) A-Q, B-S, C-R, D-P

(3) A-S, B-Q, C-P, D-R

(4) A-P, B-R, C-S, D-Q



52. Match the columns I and II and choose the correct option.

Column I		Column II	
A	Neutrophils	P	Involved in inflammatory reactions
B	Basophils	Q	Responsible for immune responses of the body
C	Eosinophils	R	Phagocytic in nature
D	Lymphocytes	S	Associated with allergic reactions

- (1) A-P, B-Q, C-R, D-S
- (2) A-P, B-R, C-S, D-Q
- (3) A-R, B-P, C-S, D-Q
- (4) A-Q, B-R, C-S, D-P



53. Statement A : Cardiac arrest means when the heart muscle is suddenly damaged by an inadequate blood supply.

Statement B : Heart failure means the state of heart when it is not pumping blood effectively enough to meet the needs of the body.

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



54. Humans have double circulation. It means

- (1) Blood flows in both side of heart
- (2) Blood is of two types- oxygenated and deoxygenated
- (3) There has two important chambers- atrias and ventricles
- (4) Blood circulates twice through the heart.



55. Statement A : A muscular chambered heart is present in all vertebrates. Fishes have 2-chambered heart with an atrium and a ventricle.

Statement B : Amphibians and all reptiles including crocodiles have a 3-chambered heart with 2 atria and a single ventricle.

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



56. Which one of the following statements is correct regarding blood pressure?

- (1) 130/90 mmHg is considered high and requires treatment.
- (2) 100/55 mmHg is considered an ideal blood pressure.
- (3) 105/55 mmHg makes one very active.
- (4) 190/110 mmHg may harm vital organs like brain and



57. If due to some injury the chordae tendinae of the tricuspid valve of the human heart is partially nonfunctional what will be the immediate effect?

- (1) The pacemaker will stop working
- (2) The blood will tend to flow back into the left atrium
- (3) The flow of blood into the aorta will be slowed down
- (4) The flow of blood into the pulmonary artery will be reduced



58. Statement A : In fishes, the gills pump out deoxygenated blood which is oxygenated by the heart and supplied to the body parts from where deoxygenated blood is returned to gills.

Statement B : In incomplete double circulation, the left atrium receives oxygenated blood from gills or lungs and right atrium gets the deoxygenated blood from the other body parts. They get mixed up in the single ventricle.

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



59. Statement A : Neural signals through the sympathetic nerves can decrease the rate of heartbeat, the strength of ventricular contraction.

Statement B : Parasympathetic neural signals increase the rate of heartbeat, speed of conduction of action potential and thereby the cardiac output.

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



60. Following are the stages of pathway for conduction of an action potential through the heart

A. AV bundle

B. Purkinje fibres

C. AV node

D. Bundle branches

E. SA node

Choose the correct sequence of pathway from the options given below

(1) E-C-A-D-B

(2) A-E-C-B-D

(3) B-D-E-C-A

(4) E-A-D-B-C



61. Identify the correct option from the following:

1) Simple organisms like sponges and coelenterates circulate water from their surroundings through their body cavities to facilitate the cells to exchange substances.

2) Different groups of animals have evolved the different method for transport.

(1) 1 is correct

(2) 2 is correct

(3) Both are correct

(4) Both are incorrect



62. Statement A : RBCs are devoid of nucleus in most of the mammals and are biconcave in shape.

Statement B : RBCs have red coloured, iron containing complex protein called hemoglobin, which play significant role in transport of respiratory gases.

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



63. Statement A : Lub is associated with the closure of the tricuspid and bicuspid valve whereas dub is associated with the closure of the semilunar valves.

Statement B : ECG is a graphical representation of the electrical activity of the lungs during a cardiac cycle.

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



64. Statement A : Rh incompatibility could be fatal to the foetus or could cause severe anaemia and jaundice to the baby

Statement B : Erythroblastosis foetalis can be avoided by administering anti-Rh antibodies to the mother immediately after the delivery of the first child.

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



65. Which one of the following blood cells is involved in antibody production.

- (1) B-Lymphocytes
- (2) T-Lymphocytes
- (3) RBC
- (4) Neutrophils



66. Statement A : Blood is a special connective tissue consisting of a fluid matrix, plasma and formed elements.

Statement B : Plasma is a straw coloured, viscous fluid constituting nearly 55% of the blood.

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



67. Agranulocytes responsible for immune response of the body are

- (1) Basophils
- (2) Neutrophils
- (3) Eosinophils
- (4) Lymphocytes



68. Blood is a special _ A _ tissue that consists of B.

- (1) A-Connective, B - Fluid matrix, plasma and formed elements
- (2) A - Connective, B - Fluid matrix, lymph and formed elements
- (3) A - Connective, B - Fluid matrix, plasma and RBC
- (4) A - Epithelial, B - Fluid matrix, plasma and formed elements



69. Assertion : Cardiac output is the volume of blood pumped by left or right ventricle in one minute.

Reason : It is calculated by multiplying the heart rate by the stroke volume.

- (1) Both Assertion and Reason are correct and Reason is the correct explanation of the Assertion.
- (2) Both Assertion and Reason are correct but Reason is not the correct explanation of the Assertion.
- (3) Assertion is correct but Reason is incorrect.
- (4) Both Assertion and Reason are incorrect



70. Which of the following statements are correct?

- I. Basophils are most abundant cells of the total WBCs**
- II. Basophils secrete histamine, serotonin and heparin**
- III. Basophils are involved in inflammatory response**
- IV. Basophils have kidney shaped nucleus**
- V. Basophils are agranulocytes**

Choose the correct answer from the options given below:

- | | |
|---------------------|--------------------|
| (1) IV and V only | (2) III and V only |
| (3) II and III only | (4) I and II only |



71. Given below are two statements

Statement A : The coagulum is formed of network of threads called thrombins.

Statement B : Spleen is the graveyard of erythrocytes.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both Statement I and Statement II are incorrect
- (2) Statement I is correct but Statement II is incorrect.
- (3) Statement I is incorrect but Statement II is correct.
- (4) Both Statement I and Statement II are correct.



72. Assertion : Systemic blood circulation occurs in human beings

Reason : The purpose of this circulation is to carry oxygen and nutrients to the body tissues and to remove CO_2 and other wastes from the tissues

- (1) Both Assertion and Reason are correct and Reason is the correct explanation of the Assertion.
- (2) Both Assertion and Reason are correct but Reason is not the correct explanation of the Assertion.
- (3) Assertion is correct but Reason is incorrect.
- (4) Both Assertion and Reason are incorrect



73. Calculate the cardiac output of an Individual having 70 heart beats/ min with a stroke volume of 55ml

- (1) 3750ml
- (2) 125ml
- (3) 3850ml
- (4) None



74. Statement A : Lymph is a colourless fluid containing specialized lymphocytes which are responsible for the immune responses of the body.

Statement B : Lymph is also an important carrier for nutrients, hormones, etc. Fats are absorbed through lymph in the lacteals present in the intestinal villi.

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



75. Cardiac activity could be moderated by the autonomous neural system. Tick the correct answer:

- (1) The parasympathetic system stimulates heart rate and stroke volume
- (2) The sympathetic system stimulates heart rate and stroke volume
- (3) The parasympathetic system decreases the heart rate but increase stroke volume
- (4) The sympathetic system decreases the heart rate but increase stroke volume



76. Statement A : An injury or a trauma stimulates the tissues in the blood to release certain factors which activate the mechanism of coagulation.

Statement B : Certain factors released by the platelets at the site of injury also can initiate coagulation of blood.

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



77. Mark the pair of substances among the following which is essential for coagulation of blood.

- (1) Heparin and calcium ions
- (2) Calcium ions and platelet factors
- (3) Oxalates and citrates
- (4) Platelet factors and heparin



78. Identify the correct statement

I. The RST complex represents initiation of the ventricular contraction.

II. The contraction starts shortly after Q and marks the beginning of the systole

(1) I is correct

(2) II is correct

(3) Both are correct

(4) Both are incorrect



79. Which of the following correctly explains a phase/ event in cardiac cycle in a standard electrocardiogram?

- (1) QRS complex indicates atrial contraction.
- (2) QRS complex indicates ventricular contraction.
- (3) Time between S and T represents atrial systole.
- (4) P-wave indicates beginning of ventricular contraction.



80. Match the items given in Column I with those in Column II and select the correct option given below :

Column I

A. Tricuspid valve

B. Bicuspid valve

C. Semilunar valve

(1) A - P, B - Q, C - R

(2) A - P, B - R, C - Q

(3) A - R, B - P, C - Q

(4) A - Q, B - P, C - R

Column II

P. Between left atrium and left ventricle

Q. Between right ventricle and pulmonary artery

R. Between right atrium and right ventricle



81. Match Column I with Column II.

Choose the correct answer from the options given below :

Column I

A. P-wave

B. Q-wave

C. QRS complex

D. T-wave

(1) A-R, B-P, C-S, D-Q

(3) A-Q, B-S, C-P, D-R

Column II

P. Beginning of systole

Q. Repolarisation of ventricles

R. Depolarisation of atria

S. Depolarisation of ventricles

(2) A-S, B-R, C-Q, D-P

(4) A-P, B-Q, C-R, D-S



82. Match the terms given under Column ' A ' with their functions given under Column ' B ' and select the answer from the options given below:

Column I

A. Lymphatic system

B. Pulmonary vein

C. Thrombocytes

D. Lymphocytes

(1) A – P, B – Q, C – R, D – S

(3) A – R, B – P, C – Q, D – S

Column II

P. Carries oxygenated blood

Q. Immune response

R. To drain back the tissue fluid to the circulatory system

S. Coagulation of blood

(2) A – R, B – P, C – S, D – Q

(4) A – Q, B – P, C – R, D – S



83. Match the following columns and select the correct option.

Column I

A. Eosinophils

B. Basophils

C. Neutrophils

D. Lymphocytes

Column II

P. Immune response

Q. Phagocytosis

R. Release histaminase, destructive enzymes

S. Release granules containing histamine

(1) A - S, B - P, C - Q, D - R

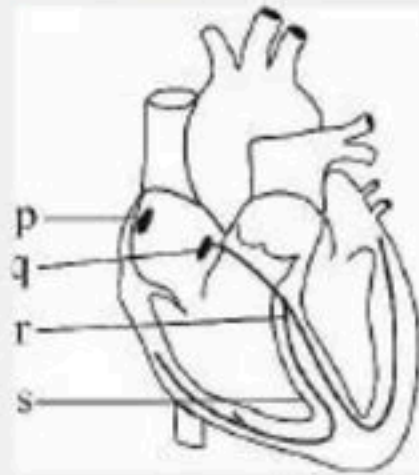
(2) A - P, B - Q, C - S, D - R

(3) A - Q, B - P, C - R, D - S

(4) A - R, B - S, C - Q, D - P



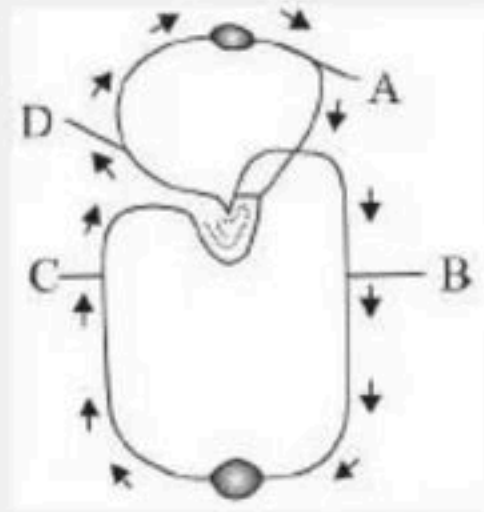
84. Identify the option showing the correct labelling for p, q, r and s with reference to conducting system of the human heart.



- (1) p – AVN, q – SAN, r – Interventricular septum, s – Bundle of His
- (2) p – Bundle of His, q – SAN, r – Interventricular septum, s – AVN
- (3) p – Interventricular septum, q – AVN, r – Bundle of His, s – SAN
- (4) p – SAN, q – AVN, r – Bundle of His, s – Interventricular septum



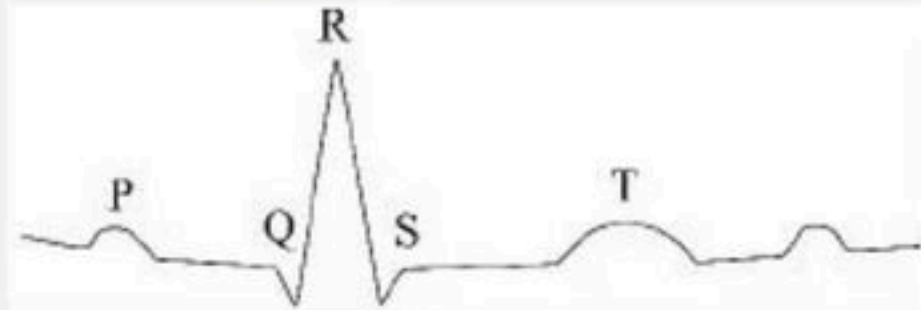
85. The given figure shows schematic plan of blood circulation in humans with label A to D. Identify the label and give its functions?



- (1) C-Vena Cava- takes blood from body parts to right auricle, $p\text{CO}_2 = 45 \text{ mmHg}$
- (2) D-Dorsal aorta - takes blood from heart to body parts, $\text{PO}_2 = 95 \text{ mmHg}$
- (3) A- Pulmonary vein - takes impure blood from body parts, $\text{PO}_2 = 60 \text{ mmHg}$
- (4) B- Pulmonary artery- takes blood from heart to lungs, $\text{PO}_2 = 90 \text{ mmHg}$



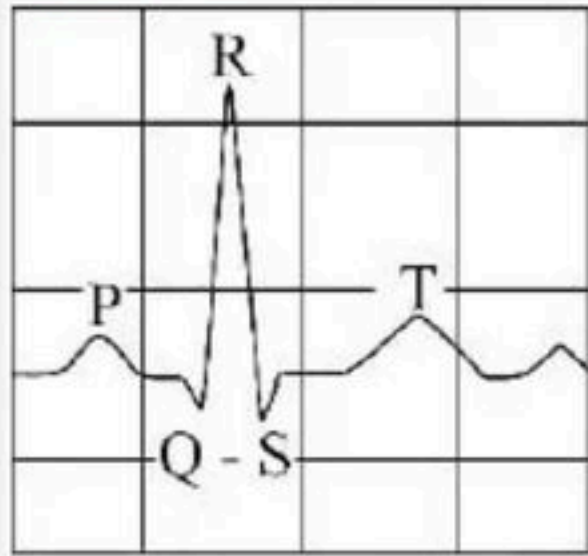
86. In the following diagrammatic representation of a standard ECG the ' T ' represents.



- (1) Depolarisation of artia
- (2) Depolarisation of ventricles
- (3) Repolarisation of artia
- (4) Repolarisation of ventricles



87. Given below is the ECG of a normal human. Which one of its components is correctly interpreted below?



- (1) Beginning of the systole
- (2) End of systole
- (3) Contraction of both the atria
- (4) Initiation of the ventricular contraction



88. Assertion : SAN initiates the heart rate

Reason : Sympathetic nerve fibres increase the cardiac activity.

- (1) Both Assertion and Reason are correct and Reason is the correct explanation of the Assertion.
- (2) Both Assertion and Reason are correct but Reason is not the correct explanation of the Assertion.
- (3) Assertion is correct but Reason is incorrect.
- (4) Both Assertion and Reason are incorrect



89. Statement A : Open circulatory system is present in annelids and chordates in which blood pumped by the heart passes through large vessels into open spaces called sinuses.

Statement B : Closed circulatory system is present in arthropods and molluscs in which the blood pumped by the heart is always circulated through a closed network of blood vessels.

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



90. Statement A : ABO grouping is based on the presence or absence of two surface antigens on the RBCs namely A and B.

Statement B : Plasma of different individuals contain two natural antibodies produced in response to antigens.

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



91. Statement A : Thrombocytes are cell fragments produced from megakaryocytes in the bone marrow.

Statement B : Platelets release a variety of factors which are involved in the coagulation of blood.

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



92. Statement A : Erythrocytes, leucocytes and platelets are collectively called formed elements and they constitute nearly 45% of the blood.

Statement B : WBCs are formed in the red bone marrow in the adults.

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



93. Statement A : Right and left atrium is separated by a thick muscular wall called inter-atrial septum.

Statement B : Left and right ventricle is separated by a thin-walled inter-ventricular septum.

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



94. Which among the followings is correct during each cardiac cycle?

- (1) The volume of blood pumped out by the Rt and Lt ventricles is same.
- (2) The volume of blood pumped out by the Rt and Lt ventricle is different
- (3) The volume of blood received by each atrium is different
- (4) The volume of blood received by the aorta and pulmonary artery is different



95. Select the incorrect statement from the following:

- I. Clot or coagulum formed mainly of a network of threads called fibrins in which living and damaged formed elements of blood are trapped**
- II. Inactive fibrinogen is converted to fibrin by the hormone thrombinase.**

- (1) I is correct only
- (2) II is correct only
- (3) Both are correct
- (4) None are correct



96. Assertion : First heart sound is 'dubb' while second heart sound is 'lubb'.

Reason : 'Lubb' is due to closing of auriculo ventricular valves while 'dupp' is due to closing of semilunar valves.

- (1) Both Assertion and Reason are correct and Reason is the correct explanation of the Assertion.
- (2) Both Assertion and Reason are correct but Reason is not the correct explanation of the Assertion.
- (3) Assertion is correct but Reason is incorrect.
- (4) Both Assertion and Reason are incorrect



97. P-wave represents

- (1) Polarization of ventricles
- (2) Repolarization of atria
- (3) Repolarization of ventricle
- (4) Depolarization of atria



98. An artery is

- (1) Thick walled in which blood flows under low pressure.
- (2) Thin walled in which blood flows under high pressure.
- (3) Thin walled in which blood flows under low pressure.
- (4) Thick walled in which blood flows under high pressure.



99. The chordae tendinae are attached to which of the following

- (1) Right AV valve
- (2) Semilunar valve
- (3) Left AV valve
- (4) 1 and 3



100. Bundle of His is a group of _____ and found in _____.

- (1) Nerve fibres, left atrium
- (2) Nerve fibres, interventricular septum
- (3) Muscular fibres, interventricular septum
- (4) Muscular fibres, left atrium



THANK YOU



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